

Few-shot object detection and attribute recognition from construction site images for improved field compliance

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ABSTRACT

Computer vision techniques can be used to detect site objects for identifying noncompliances that could lead to safety incidents. However, existing methods are limited in covering diverse hazard scenarios, detecting site objects with imbalanced distributions, and recognizing their intricate attributes to describe their conditions and functionality. To address these gaps, this paper proposes a deep learning-based method for identifying multiple fall-related objects and their associated attributes. The proposed method consists of three submethods: (1) a method for developing relevant datasets by retrieving images from open resources; (2) a method for few-shot object detection, which deals with imbalanced distributions; and (3) a method for attribute recognition to add semantic descriptions to the detected objects. The proposed method achieved an average precision and recall of 88.2% and 79.5% for few-shot object detection and 94.8% and 95.7% for attribute recognition, respectively, which indicates good performance.

1. Introduction

Improving safety on construction sites has been a longstanding major challenge. Fatalities and injuries not only cause damage to workers' health, but also result in large economic losses. Although not every noncompliance leads to an accident, a large portion of those fatalities or injuries, including those caused by fall, can be attributed to field noncompliances with safety regulations, which can be divided into two categories: noncompliant behaviors and environmental factors. For example, studies have identified that the most frequent noncompliant behaviors included failure of using personal protective equipment (PPE) and failure to follow safety procedures [1], while the most frequent noncompliant environmental factors included no available PPEs and inoperative or inappropriate equipment [2]. Construction safety regulations, on the other hand, provide detailed requirements on the existence and properties of fall protection with respect to different scenarios involving equipment, facilities, and operations. Therefore, ensuring field compliance with relevant safety regulations has been attracting an increasing amount of research attention recently.

However, it is difficult for workers to maintain compliance consistently for two reasons. On one hand, existing safety inspections are limited in their efficiency. This is because current site practices require a "competent person" to manually monitor workers to identify possible

violations. Such manual and exhaustive process is time consuming, resource intensive, and expensive [3], given the large number of site objects to be checked and thus the quantity of inspection work needed. On the other hand, and as a result, current practices rely mainly on workers' own judgement, often subject to their education levels, experiences, interpretations, awareness, safety culture, tiredness, and attitudes [4,5], which cannot guarantee that compliance is met consistently over time. For example, workers could be blind to violations due to the repeated nature of their work. Also, under production pressure, they tend to change work patterns, take more shortcuts, or use inappropriate tools or equipment [6]. Therefore, frequent safety inspection or monitoring needs to be carried out, without relying on pure manpower, to produce prompt and consistent compliance checking results.

New solutions, therefore, have been proposed towards this end by a number of research efforts, among which computer vision techniques have received high attention [7]. Object detection, in particular, provides an opportunity to automatically detect various objects involved onsite for understanding site scenes. Therefore, a growing number of research efforts in the construction safety domain have proposed different deep learning-based object detection methods and frameworks. For example, various convolutional neural network (CNN)-based models have been developed for detecting different protective equipment [3,8]. Existing research efforts have also incorporated different features or

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techniques into their detection pipeline to further enhance the performance of their deep learning-based detection methods. For example, (1) transfer learning techniques were used to leverage features extracted from images for other general applications [9]; and (2) data augmentation techniques were frequently used to overcome the problem of lacking data diversity in small training datasets [10].

However, despite the importance of these efforts, three main limitations still exist, which could negatively impact the practicality of continuously detecting and monitoring personnel, materials, equipment, and facilities onsite, through the use of computer vision techniques. First, existing research efforts are limited in the coverage of hazard scenarios and detection of granular objects, which limits the generalizability, scope, and effectiveness of these methods in field compliance checking. Second, existing detection methods often assume a balanced distribution of objects, which may not necessarily hold in real applications. In fact, some site objects can appear less frequently than others, depending on the specific activities, construction phases, or types of construction projects. Third, existing methods are often limited in recognizing the attributes of the detected objects. Safety regulations typically not only require the existence of certain protections, but also have requirements on their conditions and connections. So, detecting the attributes that are needed for field compliance checking is essential. For example, to check the compliance of ladders with the following Occupational Safety and Health Administration (OSHA) clause [1926.1053(b)(6)], more advanced image analytics are needed to recognize attributes such as “stable” or “not_stable”: “ladders shall be used only on stable and level surfaces”.

To address these limitations, this study proposes a deep learning-based method to detect fall-related site objects and their associated attributes to better capture site conditions for supporting field compliance checking. The proposed method consists of three submethods: (1) a dataset development method to retrieve and annotate a dataset that covers fall-related site objects in different scenarios; (2) a deep learning-based few-shot object detection method to address the imbalanced distribution of the objects; and (3) a deep learning-based attribute recognition method to add the semantic descriptions (to the detected objects) that are needed to check compliance with construction safety regulations. The proposed few-shot object detection method and attribute recognition method were tested and evaluated using the dataset developed using submethod 1, which consists of 1,098 images with 39 object classes and 21 attribute classes.

2. Background

2.1. Object detection

Object detection aims to recognize and localize instances of semantic objects from images or videos. It has been widely adopted in applications such as face recognition, autonomous driving, video surveillance, and anomaly detection. Depending on the process to produce predictions, deep learning-based object detection methods can be divided into two main categories: two- and one-stage detectors. Two-stage detectors usually produce a large number of candidate boxes as proposed objects first, then classify these proposals. One-stage detectors, on the other hand, use one model to perform both bounding box prediction and classification at the same time. Since less processing is needed for one-stage detectors, they typically require less computational resources and produce faster detection results compared to two-stage detectors, but their performance can be slightly compromised. In recent years, the transformer [11], which uses an encoder block and a decoder block with self-attention mechanisms to capture dependencies and relations between features, has become a dominant deep learning model structure in the natural language processing domain. In the computer vision domain, the vision transformer (ViT) proposed by Dosovitskiy et al. [12] has demonstrated better performance and less demand for computational resources than CNN-based models on large-size image classification

benchmarks such as ImageNet [13]. The detection transformer (DETR) proposed by Carion et al. [14] has shown good improvement in detecting objects without a set of hand-crafted components in the detection pipeline such as anchor generation and non-maximum suppression.

Few-shot learning is an emerging branch of deep learning-based methods that aims to achieve good performance for classes that contain only a few training samples (known as novel classes), using the knowledge learned from classes that are more frequent (known as base classes) [15]. Compared with regular deep learning approaches that rely on large amounts of training data, few-shot learning-based approaches can better capture the underlying patterns and grasp the essential connections between new concepts and the existing knowledge, thus can generalize to new tasks that usually contain only a few labeled samples [sometimes even one sample (known as 1-shot learning) or zero samples (known as 0-shot learning)] [16]. The idea of few-shot learning was borrowed from the fact that even toddlers can easily learn a new concept with limited information given to them. Such learning and generalizability is especially useful for applications in the construction safety domain, because of the difficulty in collecting and annotating a large number of site images, as well as the highly imbalanced distribution of site objects. There are two main types of few-shot learning methods: meta and metric. Meta learning-based methods adapt to the task of predicting novel classes with the help of task-level meta knowledge such as the structure and variability of the dataset. Metric learning-based methods, on the other hand, learn to compare data samples to the support samples (a subset of the images with K examples for each object class) in terms of their similarity to distinguish object classes. Few-shot object detection has been attracting an increasing amount of research attention. For example, a number of research efforts in the healthcare domain have also developed few-shot object detection methods to improve the identification of rare diseases (e.g., [17]), where such cases occur less frequently and a ground truth is difficult to obtain.

2.2. Attribute recognition

Attribute recognition aims to identify a group of attributes for each object as semantic descriptors. It is an emerging field for various surveillance applications such as face verification and pedestrian intention prediction. For example, research efforts in the computer vision domain have predefined attributes either focused on physical appearance of people such as gender, age, hair length, and clothing [18], or human actions and motions such as talking and pushing [19]. The identified attributes could provide fine-grained information in terms people's current behavior (e.g., talking on the phone) with respect to gender and race, which could be used in subsequent pedestrian intention analysis (e.g., predicting if the pedestrian is going to cross the road or not) [20–23]. In the construction safety context, current conditions of workers, equipment, machines, and their environment, in terms of functionality, occupancy, motion, and shape, can be continuously changing, which need to be frequently checked to ensure compliance over time. Therefore, more detailed attributes as required by applicable construction safety regulations need to be recognized from site images to describe those conditions, which can provide more fine-grained semantic descriptions for the detected site objects and support subsequent compliance analysis.

Existing attribute recognition methods can be divided into three categories: relation-, attention-, and part-based methods. Relation-based methods utilize interdependency and correlations among attributes to aid the attribute recognition process [24]. However, such methods usually require predefined attribute relations that are obtained through external linguistic knowledge bases, which do not exist in the construction domain and are thus not practical or scalable. Attention-based methods utilize visual attention mechanisms to learn the distinctive features of each attribute [25,26]. Improved accuracy can be obtained for such methods, compared to the relation-based ones. Part-based

methods learn to decompose an image into a set of patches, extract salient patterns from localized patches, and fuse these patterns for a distributed representation [27]. This type of methods is more promising due to its ability to learn not only the distinctive features of each attribute, but also the regions where each attribute should be located. In the construction safety domain, this type of methods can provide a robust solution to recognize attributes of site objects, despite that the location of their features indicating attributes are constantly moving or changing due to the dynamic nature of construction sites.

3. State of the art and knowledge gaps

Computer vision techniques have been investigated for various applications in the construction safety domain, which pave a solid foundation for detecting fall-protection noncompliances due to unsafe behaviors and hazardous work conditions. For example, Cheng et al. [28] and Yan et al. [29] focused on detecting and tracking labor and equipment such as barricades, ladders, and PPEs. Tang and Golparvar-Fard [30] and Tang et al. [31] focused on estimating and monitoring workers' behaviors such as poses, actions, and motions. Focusing on fall protection specifically, a body of literature proposed different object detection methods. For example, Khan et al. [32] detected PPEs for workers on scaffolds, with the help of Internet-of-Things (IoT) technologies. Wang et al. [33] focused on detecting PPEs such as safety vests when working at heights for measuring safety compliance. Fang et al. [34] detected harnesses of workers at heights to proactively identify unsafe behaviors.

Despite the large number of efforts in the construction safety domain, there still exist three main challenges in detecting site objects and understanding their conditions for improved construction safety. First, existing research efforts usually lack in the coverage of hazard scenarios that can involve multiple site objects, object components, object appearances, and site scenes, which can limit the generalizability, scope, and thus the effectiveness of these methods in field compliance checking. Existing efforts are also limited in detecting granular objects from site images. They (e.g., [35]) typically detect high-level objects (e.g., scaffolds), but do not detect granular objects such as equipment components (e.g., toeboard, guardrail, etc.). Detecting equipment and equipment components (and associating them) is important for supporting safety compliance checking, which is in many cases related to the conditions and use of the equipment and their components. Therefore, this paper seeks to develop enhanced image retrieval and data preprocessing techniques to leverage online resources to cover diverse hazard scenarios (as covered in the relevant safety requirements) in the collected images.

Second, existing detection methods typically assume a balanced distribution of all object classes, which does not necessarily hold in real practice. In fact, some object classes (e.g., holes) can appear less frequently but can have such a high impact on compliance that if neglected, severe accidents can happen. Therefore, it is necessary to develop a method that can address the imbalanced distribution of all object classes and still achieve good performance in identifying object classes even when scarce. Existing research efforts have adopted different techniques such as data augmentation and upsampling and/or downsampling to address this problem on a dataset level. For example, Bang et al. [36] proposed a data augmentation method to address class-distribution imbalance and diversify the patterns in the dataset of construction resources using techniques such as removing-and-inpainting, cut-and-paste, and image transformation. Chen et al. [10] proposed a context-guided data augmentation-based method that utilizes location information of construction resources as priors to generate artificial samples with occlusions, which compensates for the lack of occluded images for collision prevention. However, those methods usually require a set of hand-crafted rules as guidance to conduct transformation or generate new features, which were typically defined by a domain expert and could be specific to the dataset [37]. Thus, the generalizability of

those methods could be negatively affected. Only recently, a number of research efforts in the construction domain have taken on a few-shot learning approach to adapt to novel classes. For example, Kim and Chi [38] proposed a meta learning-based few-shot learning method to detect five construction heavy machines such as excavators and loaders, which showed better results in detecting classes that contain only 30 images than regular supervised learning methods. Tran-Anh et al. [39] proposed a cross-feature and attentive-squeeze network to conduct few-shot detection of landslides from satellite images, when such disasters do not occur on every part of a road. Khoshboresh-Masouleh and Shah-Hosseini [40] proposed a CNN-based few-shot method to detect building changes from remote sensing images because the cost to obtain high quality images from different countries/regions can vary significantly, which showed high robustness and better coverage when detecting changes from areas with small training data. However, in terms of features, the aforementioned efforts did not take relative location information of different site objects into account, especially the position of an object in the image or connections between one equipment and its components, thus were limited in efficiently utilizing the hidden features from site images. In terms of model architecture, those few-shot learning methods were mainly developed based on CNN models rather than transformers, which show better representation ability of objects through the use of embedding components [41]. In terms of machine learning approaches, most of those efforts are mainly fully supervised methods. There is a lack of efforts using unsupervised or semi-supervised backbone models, which could further reduce the amount of annotated data. Therefore, this paper seeks to develop a transformer-based few-shot detection method that considers the relative location information of the objects, with a certain level of unsupervision, for detecting objects with varying occurrences using limited amount of data.

Third, existing research efforts mainly focused on detecting the existence of certain protections, without extracting visual information that describe their functionality, occupancy, and deflection, which are essential information that must be captured and checked to ensure field compliance. For example, to check compliance with the OSHA clause 1926.451(b)(1), which requires "scaffolds shall be fully planked", it is important to extract information on whether the scaffold is fully planked or not fully planked. Existing efforts are limited in extracting such an attribute that describes the condition of the scaffold in the same way (in terms of meaning and terminology) as described in the OSHA regulations. Attributes studied in previous efforts were mainly limited to elemental attributes such as color, geometry, and auxiliary information. For example, Fang et al. [42] and Fang et al. [43] considered attributes such as class, distance, color, height, speed, and coordinates. Chen et al. [44] considered a set of attributes such as "hasName", "hasDescription", and "hasRequirementType" to integrate linguistic and visual information for on-site hazard identification. However, such attributes are not sufficient in describing the detailed and constantly changing conditions of site objects to ensure consistent compliance over time. This is because construction safety regulations provide requirements not only related to the existence, types, sizes, or distances of certain protections (as could be indicated by class, color, and coordinates for example), but also with regard to intricate attributes such as service, functionality, occupancy, impairment, quality, and capacity of the protective measure (e.g., "Unstable objects shall not be used to support scaffolds or platform units."), which could be equally, if not more, important to recognize for two reasons. First, intricate attributes often serve as the main criteria for both selecting applicable protections and checking their compliance. Second, those intricate attributes about the conditions of the objects could be changing over time due to the dynamic nature of construction sites, while color and geometry information are more likely to be fixed. However, recognizing intricate attributes can be technically challenging. First, recognizing such attributes accurately from site images often requires interpretation and/or inference (e.g., infer the potential motion of an object to describe it as "not_stable"), which requires identifying high-level semantic information, implicit correlation, and

causal relation. Second, intricate attributes can vary in their scales and detail levels, which could require extracting feature maps at multiple levels and allocating different amounts of attention accordingly. For example, whether a scaffold is fixed to a wall is a finer attribute than whether the scaffold is level or not, which can require more visual details and thus higher attention. Third, it is necessary to develop methods that can learn to relate attributes to some discriminative regions because attributes can often occur in some particular regions. However, it is difficult to develop such a method without relying on a set of predefined rules or additional region annotations, which requires a large amount of manual work and is thus expensive and time consuming. Therefore, this paper seeks to develop an attention-based attribute detection method to recognize multiple attributes of the detected objects by associating attributes to particular regions, without additional rules or annotations.

4. Proposed method

This study proposes a deep learning-based method to detect fall-related site objects and their associated intricate attributes, for capturing essential site condition information to check compliance with relevant safety regulations, particularly OSHA. The proposed method consists of three main submethods, as per Fig. 1: (1) a dataset development method to retrieve and annotate fall-related site images, (2) a few-shot object detection method to address the imbalanced distribution of the objects, and (3) an attribute recognition method to add semantic descriptions (to the detected objects) that are needed to check compliance with construction safety regulations.

The dataset development method aims to retrieve, annotate, and augment fall-related site images. A query-based method with augmentation techniques was used to retrieve and develop site images of different fall-hazard scenarios. The dataset development method consists of three steps: image retrieval, annotation development, and data augmentation. Image retrieval aims to query and crawl real-life construction site images from multiple open image resources. Annotation development aims to add labels to the objects in the collected images. Data augmentation helps add richness and variations in features to the dataset by adding slightly modified copies of the existing images. Few-shot object detection aims to identify various fall-related site objects, especially less frequent object classes, by using the knowledge learned

from more frequent object classes. A transformer-based method with a semi-supervised region proposal generator was developed, which can better capture the dependencies and relations between visual features with fewer annotated examples. A total of 39 fall protection-related object classes (see Fig. 2) were considered, based on fall-related topics from OSHA and entity names extracted in [45], to ensure that the same semantics are maintained across both safety requirements and the detected information for supporting subsequent automated reasoning. Attribute recognition aims to add semantic descriptions to the detected objects, in terms of their conditions, to provide supporting information for subsequent compliance comparisons. A part-based method with spatial attention localization modules was developed, which can locate attributes to specific regions and allocate higher attention to more fine-grained attributes. A total of 21 attribute classes were considered, based on the property and attribute names, which were extracted from OSHA regulations in [45]. The attributes, along with examples of OSHA clauses that refer to these attributes, are shown in Table 1. For example, OSHA clause 1926.451(f)(4) states “any part of a scaffold damaged or weakened shall be immediately repaired or replaced”. Hence the attribute “damaged” was extracted. For more details on how the objects and attributes were extracted from OSHA, the readers are referred to Wang and El-Gohary [45,46]. Fig. 3 shows an example that illustrates the final output of the proposed method (i.e., detected objects and attributes), examples of the applicable OSHA clauses in this scenario, and how the detected site condition information can be aligned with safety regulations for subsequent compliance checking.

4.1. Dataset development

4.1.1. Image retrieval

Image retrieval in this study aims to collect a number of site images from multiple online open image resources. Two main online sources were selected for retrieving images: Flickr and Google Images. Flickr is an image management and sharing application or community that contains more than 5 billion images, each with detailed tags such as people and buildings to indicate its content. The Flickr application programming interface (API) was used to retrieve images from Flickr's construction-related image groups. The Flickr API allows users to access Flickr's image collection from its website, through constructing an

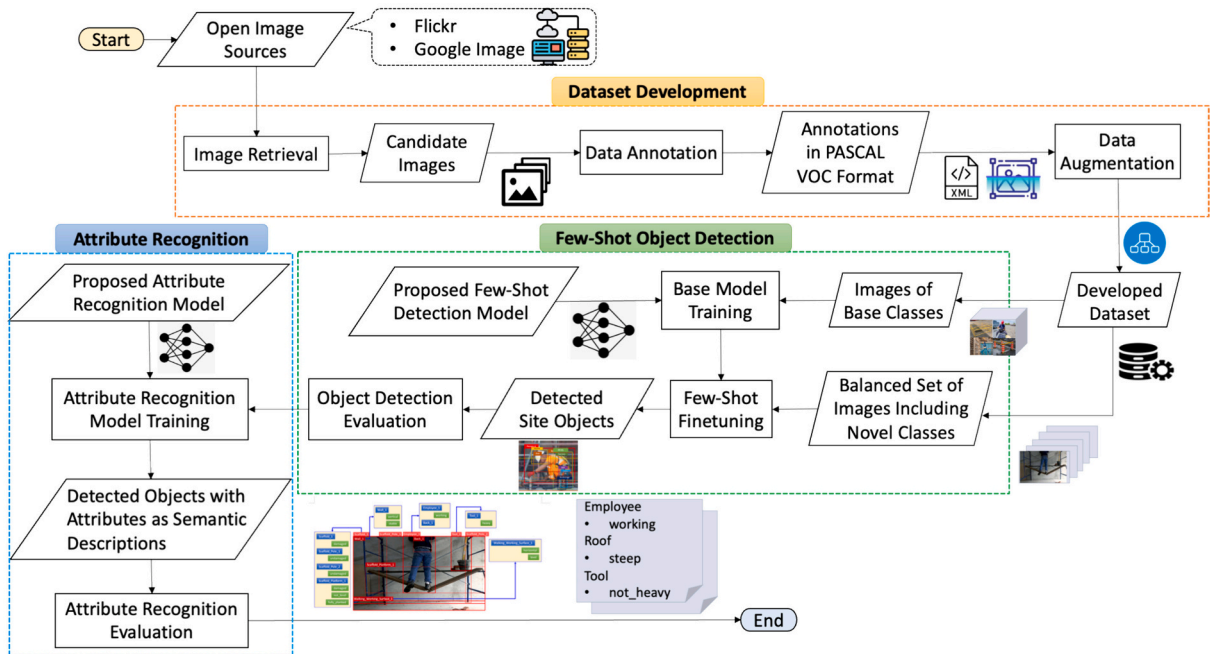


Fig. 1. Research methodology.

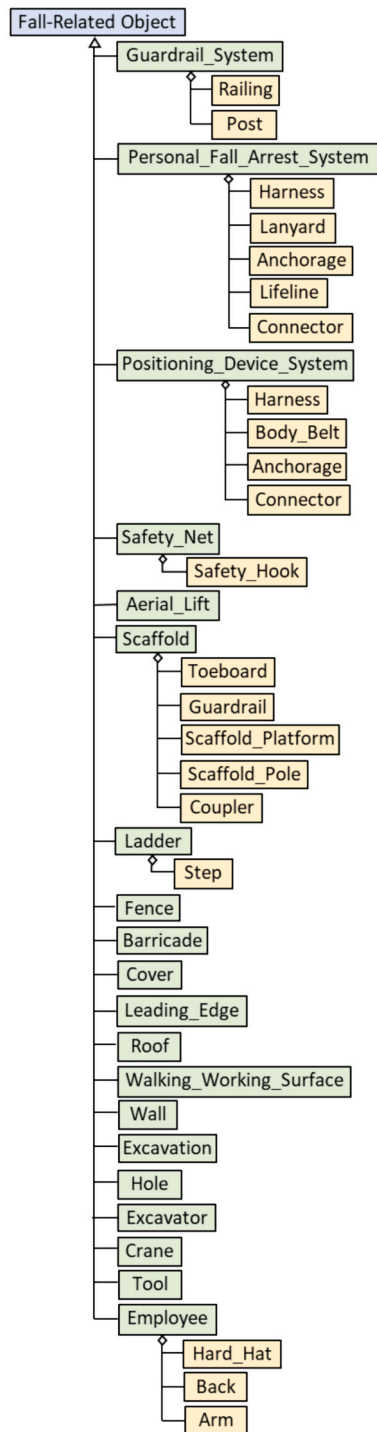


Fig. 2. Fall protection-related object classes.

endpoint (often in the form of a web address), which returns data in the format specified by users. To retrieve images more specific to fall protection, keywords for different fall-related topics were used as queries. To retrieve as many candidate images as possible, keywords and their synonyms were used. Candidate images from Google Image were also retrieved to add more samples for developing the dataset. To filter out some irrelevant images, queries were designed by concatenating words such as “construction” and/or “safety” to the original keywords, for restricting the application domain. After retrieving a set of candidate images, three types of images were removed to improve the quality of the dataset: (1) images about other accident types such as struck-by or

Table 1

Attribute Classes.

Attribute class	Examples	Example applicable OSHA clauses
Damaged	If the scaffold boards are damaged	1926.451(f)(4),
Undamaged	or undamaged	1926.502(c)(5)
Vertical	If the walking/working surface is vertical or horizontal	1926.502(d)(9),
Horizontal		1926.502(d)(6)(iv)
Steep	If the roof is steep or low-sloped	1926.501(b)(11),
Low_sloped		1926.501(b)(10)
Level	If the ground for supporting scaffold is level or not level	1926.451(c)(2)(i),
Not_level		1926.1053(b)(6)
Stable	If the wall under construction is stable or not stable	1926.451(c)(2)(ii),
Not_stable		1926.451(c)(2)(iii)
Protected	If the leading edge is protected or not protected	1926.501(b)(1),
Not_protected		1926.501(b)(10)
Fixed	If the anchorage is fixed or not fixed	1926.451(g)(3)(i),
Not_fixed		1926.1053(a)(1)(iii)
Heavy	If a tool carried by the worker is heavy or not heavy	1926.451(h)(1),
Not_heavy		1926.1053(b)(22)
Walking	If the worker is walking, working, or standing on the surface	1926.501(b)(14),
Standing		1926.453(b)(2)(iii)
Fully_planked	If the scaffold platform is fully planked or not fully planked	1926.451(b)(1),
Not_fully_planked		1926.451(b)(1)(ii)

caught-in-between; (2) images of poor quality to the extent that objects can be difficult for humans to perceive. For example, some candidate images were taken so far away that workers were too small to be perceived. Similarly, some images may have very low resolution, be too blurry or dark, or contain simulated cartoons that are unrealistically rendered; and (3) fisheye images and repeated images.

4.1.2. Data annotation

Data annotation is the process of adding labels to the objects in the collected images, following a labeling scheme. For few-shot detection, the labeling scheme of the PASCAL VOC dataset [47] was followed in this study using an annotation tool named LabelImg. Labels for each object instance contain two kinds of information: object class name and object coordinates. Labels for each image containing multiple objects were stored in an (Extensible Markup Language) XML file, which is a common markup language format for storing structured data. For attribute recognition, the labeling scheme of the PEdesTrian Attribute (PETA) dataset [48] was followed, which requires adding multiple attribute tags to an object in its existing XML file. Those multi-label attributes were then converted into one-hot encoding as the input for attribute recognition.

4.1.3. Data augmentation for creating image variations

Data augmentation in this study aims to create image variations by adding slightly modified copies of the existing data through small transformations such as geometric transformations and color space augmentations [49], such that the characteristics of the dataset (e.g., object scales) could resemble the true variations/distribution of site images. With those augmented training samples, the deep learning model would traverse over a larger feature space, better capture and extract the underlying patterns, and thus could be more robust to small changes, dislocations, and transformations. These variations can also act as a regularizer to prevent overfitting, which is important in further ensuring the generalizability of the proposed method. The following data augmentation techniques were used: random color jittering, random jpeg compression, random grayscale conversion, random flipping, and random rescaling [49]. Random rotation was not included because angles are important in determining certain attributes. For example, whether a roof object can be recognized as a steep roof or low-sloped roof can depend on the angle of the roof panels.

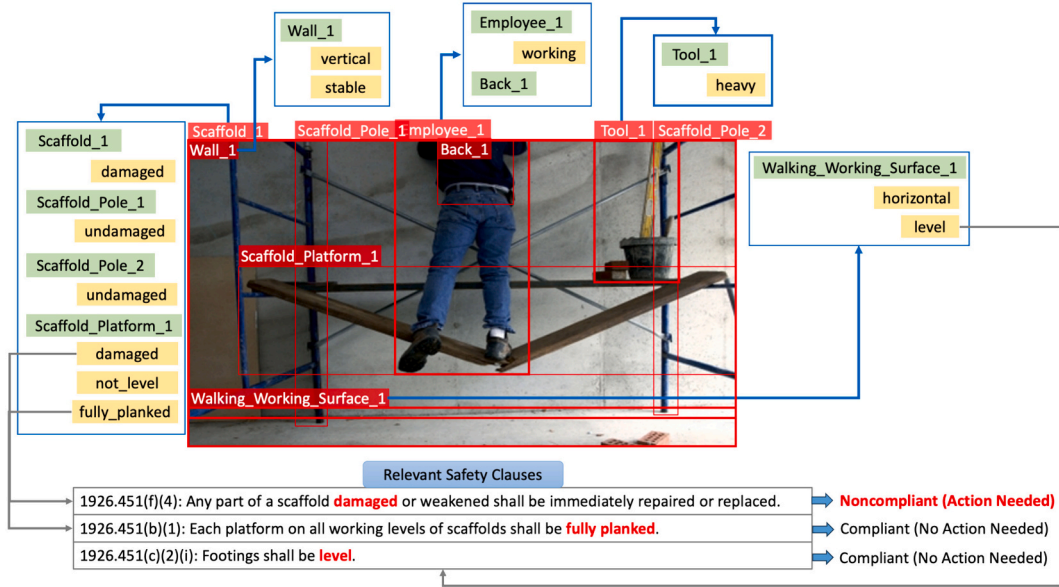


Fig. 3. Example to illustrate proposed method (output, applicable clauses, compliance checking).

4.2. Few-shot object detection

Few-shot object detection aims to develop an object detection model that can generalize to novel objects with limited site images, in order to address the imbalanced distribution of site objects. The proposed method is a finetuning method that combines the benefits of both meta- and metric-learning, as described in Section 2.1.2. The few-shot detection method consists of two steps: base model training and few-shot finetuning. Base model training focuses on extracting features from base classes where there exist abundant training samples. It was developed based on a transformer model with position encoding and a semi-supervised region proposal generator that can extract composition features. During few-shot finetuning, the model adapts to the task of detecting novel classes where there exists limited number of training samples, by utilizing the knowledge learned during base model training.

4.2.1. Base model training

The proposed base model contains three main components: a semi-supervised region proposal generator, an encoder-decoder transformer, and several feed forward networks (FFN) to produce predictions. The semi-supervised region proposal generator further consists of: (1) an unsupervised backbone model, swapping assignments between multiple views of the same image (SwAV) [50]: it seeks to map features to their embedding space in a self-supervised manner, which helps further reduce the amount of annotated data needed. It applies a set of transformations to augment an input image with different views and predicts the cluster assignments of these images as pseudo-labels to learn visual representations; and (2) a supervised selective search process [51]: it seeks to capture the connectivity between one equipment and its components. It utilizes region-based features in a bottom-up manner when generating region proposals, by generating patches of object regions at different scales and iteratively grouping regions that should be connected. To select a variety of object proposals with high probabilities, pseudo ground-truth region proposals (b) were then generated from such composition features through importance sampling. Object scores (c) for all pseudo ground-truth region proposals were assigned as 1.

The encoder-decoder transformer aims to extract object features and localize the objects from an input image by encoding and decoding their visual features, with special attention on their position information. The encoder block takes in the output from the semi-supervised region proposal generator and generates object embeddings (emb). To preserve

the relative location of an object in an input image, two-dimensional positional encoding of the relative height and width for each object was added as input to the encoder block. This is similar to the relative positional embedding from the computational linguistics domain [52], where positional encoding is typically used to preserve the relative location and thus the importance of a word token in a sentence, which helps achieve better generalizability.

The decoder block, on the other hand, seeks to decode the object embeddings from the encoder and calculate object class predictions. It consists of: (1) a multi-head self- and a multi-head decoder-encoder attention modules, which take in an input image as well as the learned positional encodings and computes N embeddings of size d for N objects in parallel; (2) an FFN, which computes predictions of the bounding box coordinates ($\hat{b}$) for the input image. It also produces estimated object embeddings ($\hat{emb}$), which are to be compared with the object embeddings (emb) from the encoder block; and (3) a linear projection layer, which calculates probabilities of object classes ($\hat{c}$) for each bounding box using a linear layer and a softmax layer. The encoder-decoder transformer architecture is illustrated in Fig. 4.

The bipartite matching algorithm [14] was used to measure the alignment between the pseudo ground truth (i.e., emb , b , c) and the predicted results (i.e., $\hat{emb}$, $\hat{b}$, $\hat{c}$). It aims to find the permutation σ such that it minimizes the cost between ground-truth y and the estimated label $\hat{y}$, as per Eq.1:

$$\sigma = \arg \min_{\sigma \in \mathcal{O}_N} \sum_{i=1}^N L_{match}(y_i, \hat{y}_{\sigma(i)}), \quad (1)$$

where $L_{match}(y_i, \hat{y}_{\sigma(i)})$ is a pairwise loss function defined in [53] and $\mathcal{O}_N$ is the set containing all possible permutations. Training samples of base classes were used during this step with the following loss function:

$$L(y, \hat{y}) = \sum_{i=1}^N \lambda_f L_{class}(c_i, \hat{c}_{\sigma(i)}) + \mathbb{1}_{\{c_i \neq 0\}} [\lambda_b L_{box}(b_i, \hat{b}_{\sigma(i)}) + \lambda_e L_{emb}(z_i, \hat{z}_{\sigma(i)})], \quad (2)$$

where L_{class} is the cross-entropy loss for measuring class predictions, L_{box} is a linear combination of L_1 loss and generalized intersection over union (GIoU) loss [54] for measuring box predictions, and L_{emb} is the L_1 loss for measuring the distances between embeddings.

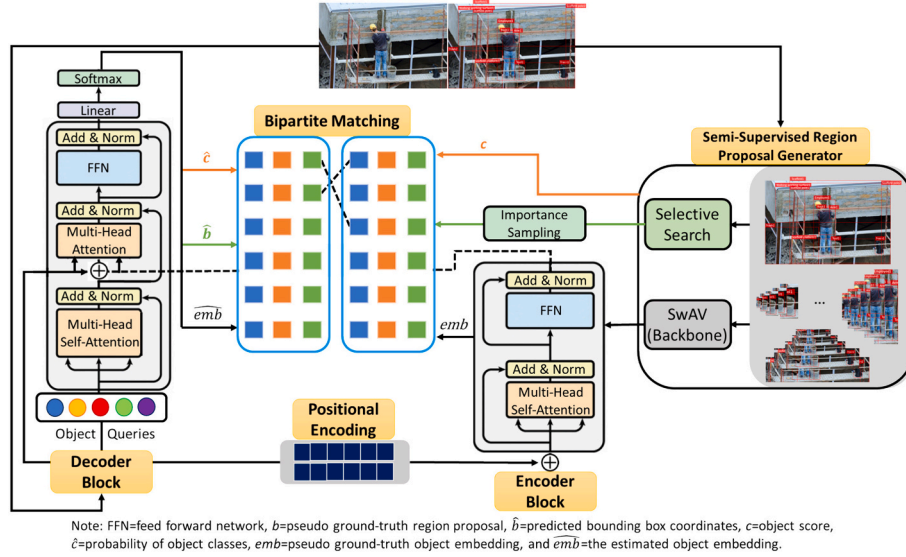


Fig. 4. Architecture of proposed method for few-shot object detection.

4.2.2. Few-shot finetuning

During few-shot finetuning, the same model architecture as in base model training was used with a different training dataset. The training dataset contains a small and balanced set of images for all object classes (both base and novel classes): the number of samples is the same across all object classes and is small in quantity (i.e., K -shots with $K = 50$ in this study). To finetune the detection model, weights of the linear projection layer were first randomly initialized for the novel classes. Then, the FFN from the decoder block and the linear projection layer were finetuned using the new training dataset, with the weights of the entire encoder block fixed. The same loss function as in Eq. 2 was used, with a smaller learning rate (1/2 of the learning rate in base model training).

4.3. Attribute recognition

A deep learning attribute recognition model was developed to identify a set of attributes for each site object to provide detailed semantic information for describing its conditions. A part-based method is proposed in this study to locate attributes to their related regions without additional region annotations and pay different amounts of attention according to the attributes' detail levels. The deep learning model proposed in this paper consists of a main batch norm-inception network [55] with feature pyramid networks (FPNs) [56], and a set of proposed spatial attention localization modules (SALMs). The batch norm-inception network can better capture the feature distribution, due to its normalization process to prevent gradient vanishing, especially for deeper networks. FPNs use a set of top-down or skip connections to extract and concatenate features at different levels. This is because low-level details and high-level semantics are often complementary in contributing to attribute prediction. Thus, the batch norm-inception network with FPNs of the model has two main paths: bottom-up and top-down pathways. The bottom-up pathway consists of multiple batch norm-inception blocks at different feature levels, which were used to extract features from input images in a bottom-up manner. The top-down pathway consists of three lateral connections and two top-down connections, as shown in Fig. 5, where the lateral connections can reduce the dimensions of bottom-up features, and the top-down connections can transmit the higher-level features through an upsampling operation. Features from adjacent levels are concatenated following Eq. 3:

$$\mathbf{X}_k = \{h(\phi_k(I)), g(\mathbf{X}_{k+1})\}, k \in \{1, 2\}, \quad (3)$$

where h denotes a convolutional layer for dimension reduction, $\phi_k(I)$ is the bottom-up features for input image I at feature level k , and g refers to an upsampling operation with nearest neighbor interpolation. Because the highest level has no top-down connections, it only contains h to be concatenated with features from other levels, i.e., $\mathbf{X}_3 = h(\phi_3(I))$.

The proposed SALMs are then used to automatically discover the discriminative part of an object by: (1) localizing the regions where each attribute can occur, and (2) allocating different attentions to attributes with different detail levels. A SALM consists of a spatial transformer layer and a channel-attention subnetwork. The spatial transformer layer is a flexible version of a spatial transformer network (STN) [57], which applies spatial transformation such as cropping, translation, and scaling to feature maps. The channel-attention subnetwork, on the other hand, is used to reweight features for different attributes according to their detail levels, because more attention needs to be paid to finer attributes. For example, "fixed" (i.e., whether a scaffold is fixed to a wall or not) is a finer attribute that requires the model to zoom into a particular area, capture more visual details, and thus pay higher attention. Each channel-attention subnetwork takes the features at each level produced by the batch norm-inception network with FPNs, and calculates channel-wise attention weights through a set of layers including global pooling, convolution, rectified linear unit (ReLU), and sigmoid. The SALM then applies spatial transformation to both the features and the attention weights. A fully connected (FC) layer is used at the end to conduct inference and calculate predictions. During inference, a deep supervision mechanism is used to provide feedback to all SALMs at different feature levels in terms of whether the distinctive regions are accurately captured. This is achieved by aggregating outputs from SALMs through a voting scheme that produces the maximum responses across all feature levels. The best predictions with the most accurate attribute regions are selected using such maximum voting scheme, which is proved to be a better ensemble method than the averaging voting scheme [58].

During training, each object was fed into the main network through both bottom-up and top-down pathways. Features extracted using the batch norm-inception network with FPNs were concatenated before being fed into the SALMs. The SALMs then performed attribute-specific localization and channel-wise attention weight calculation to produce intermediate attribute prediction vectors. Those prediction vectors were aggregated using an element-wise maximum operation for producing final predictions. The process of the proposed attribute recognition model is illustrated in Fig. 5. The loss function in the form of weighted binary cross-entropy was used at each feature level, as per Eq. 4.

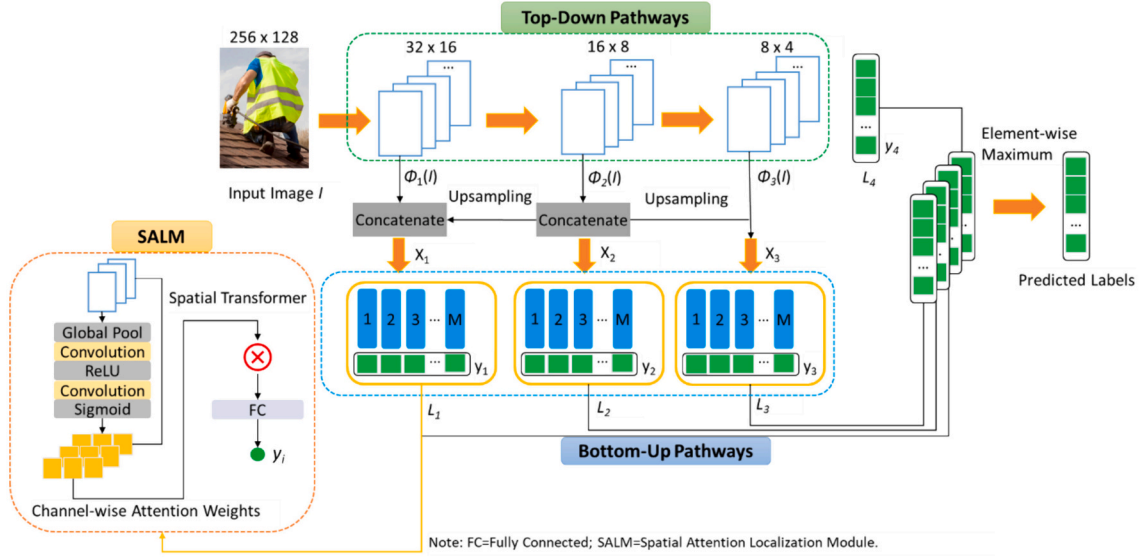


Fig. 5. Architecture of proposed model for attribute recognition.

$$L_k(\hat{y}_k, y) = -\frac{1}{M} \sum_{m=1}^M \gamma^m (y^m \log(\sigma(y_k^m)) + (1 - y^m) \log(1 - \sigma(y_k^m))), \quad (4)$$

where k is the k^{th} level, y is the ground-truth label, $\hat{y}_k$ is the predicted label from k^{th} level, M is the number of attributes, $\gamma^m = e^{-a_m}$ is the loss weight for the m^{th} attribute, e is the exponential constant, a_m is the prior class distribution of the m^{th} attribute, and σ is the sigmoid activation. The overall loss is obtained by summing all losses at the individual feature levels.

4.4. Evaluation metrics

Precision, recall, and $F-1$ measure were used for evaluating the performance of few-shot object detection and attribute recognition. As per Eqs. 5–6, precision (P) is defined as the percentage of the number of correctly detected objects (or attributes) divided by the total number of predicted objects (or attributes). Recall (R) is the percentage of the number of correctly detected objects (or attributes) divided by the total number of objects (or attributes) that should be detected. $F-1$ measure is the harmonic mean of precision and recall, as per Eq. 7. For object detection, mean average precision (mAP) was also used to evaluate the performance of the proposed method. mAP is based on average precision (AP), the area under the precision-recall curve. A larger AP indicates better performance. However, AP itself is not sufficient in indicating how the model performs in localizing object instances. The intersection over Union (IoU) is thus used to measure the overlap between a ground-truth mask and the predicted mask. For example, AP@50 means AP at an IoU overlap threshold of 50%, with mAP being the average of AP@50 values over all instances in this case. For attribute recognition, the performance was evaluated at two levels: (1) at the attribute level (i.e., average performance per attribute class); and (2) at the object level (i.e., average performance per object instance). Differences in performance across both levels could be seen due to the imbalanced distribution of attributes.

$$P = \frac{\text{number of correctly detected objects (or attributes)}}{\text{total number of predicted objects (or attributes)}} \quad (5)$$

$$R = \frac{\text{number of correctly detected objects (or attributes)}}{\text{total number of objects (or attributes) that should be detected}} \quad (6)$$

$$F-1 = \frac{2 \times P \times R}{P + R} \quad (7)$$

5. Experimental results

The proposed few-shot detection and attribute recognition method for identifying site objects and their attributes was tested using the dataset developed in Section 4.1. A set of experiments were conducted to: (1) evaluate the performance of the proposed few-shot object detection method using different splits of base/novel object classes; (2) evaluate the impact of few-shot detection by comparing its performance with a general detection method; and (3) evaluate the effectiveness of the proposed attribute recognition method by comparing the performance with and without the spatial attention localization modules. The hyperparameters of the few-shot object detection and attribute recognition models were finetuned to achieve the best performance. The experiments were conducted using a g4dn.8xlarge Amazon web services (AWS) elastic compute cloud (EC2) instance with an Ubuntu system.

5.1. Dataset characteristics

The collected dataset included a total of 1,098 images related to fall protection, which vary in their resolution, viewpoints, object scales, color histograms, lightning conditions, and camera information. For example, Fig. 6 (a) shows examples for viewing a working surface from different angles, which can encourage the model to learn a better overview of the surface. Fig. 6 (b) shows examples of excavators in different colors, which can encourage the model to capture more features other than color. Fig. 6 (c) shows examples of varying qualities, such as different exposures, resolutions, weather conditions, and lightning, which can improve the robustness of the object detection model to such variations. Fig. 6 (d) shows examples containing different coverages of the same equipment, which can encourage the model to predict based on partial cues. Fig. 6 (e) shows images containing different shapes of barricades, which can encourage the model to capture features other than geometry, such as through understanding the connections between barricades and excavations. As shown in those examples, most objects are placed in cluttered background with varying sizes.

After annotation, a total number of 12,063 object instances were obtained, which is equivalent to 11.0 object instances per image. This dataset size is comparable to a few similar research efforts in the construction safety domain, in terms of both the number of images and the

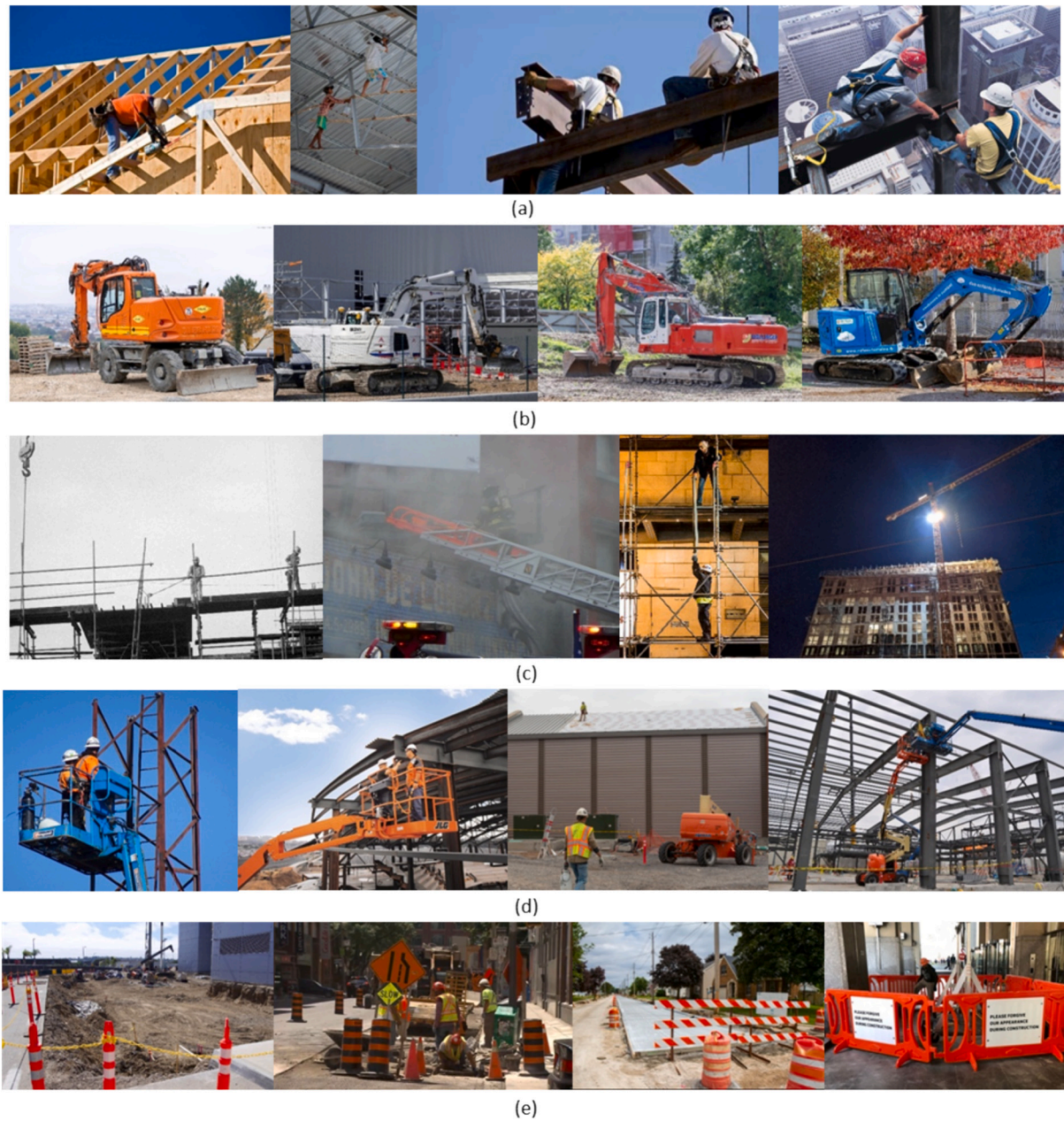


Fig. 6. Examples of variations in the dataset: (a) Different viewpoints of working surfaces; (b) Different colors of excavators; (c) Different image qualities; (d) Different coverages of equipment; and (e) Different shapes of equipment.

number of object instances. For example, 1,000 images were used for safety guardrail detection in [59], and 4,700 object instances were used in [3]. The 12,063 object instances include 43.2%, 32.6%, and 24.2% small, medium, and large objects, respectively, which is comparable in terms of scale distribution to other construction safety datasets of site images, such as the dataset by Tang et al. [31], which is composed of 42.6%, 32.9%, and 24.5% small, medium, and large objects. Object scales can be divided into small, medium, and large by the pixels they cover, where a small box covers an area of less than 32-by-32 pixels and a large box covers more than 96-by-96 pixels [31]. Looking at a different size classification to compare to general-domain datasets of unmanned aerial vehicle (UAV) images, 57.5% of the annotated objects are less than 50-by-50 pixels. According to [60], bounding box sizes of most objects (specific number not given) in UAV images are less than 50-by-50.

In terms of object classes, Fig. 7 shows the distribution of all fall-

related objects in the dataset, which reflects an imbalanced distribution of all object classes. For example, wall instances appear far more often than hole instances. However, detecting holes accurately can be of significant importance in ensuring compliance and safety, in spite of its rare occurrence. This reinforces the need to use few-shot detection rather than general detection methods for identifying such site objects.

5.2. Few-shot object detection

5.2.1. Performance of proposed few-shot object detection method

To evaluate the effectiveness of the proposed few-shot object detection method, three different splits of base/novel object classes were used, because the performance can vary due to different selections of novel classes. For each split, the 39 object classes were divided into 35 base classes and 4 novel classes. So, in total (i.e., for the three splits), 12 classes were considered as novel, with no overlapping across the splits.



Fig. 7. Distribution of object classes.

This is similar to the method of splitting in existing few-shot detection benchmarks in the computer vision domain [15], but without overlapping due to the characteristics of the developed dataset. The number of testing samples for all object classes is summarized in Table 2.

Using 50-shots for finetuning, the proposed few-shot detection method achieved an average precision, recall, and *F-1* measure of 88.2%, 79.5%, and 83.6%, respectively, for all object classes, which indicates that the proposed method was effective in extracting the target information from the site images. The proposed method also achieved an average precision, recall, and *F-1* measure of 88.6%, 80.3%, and 84.2% for the base classes, and 85.6%, 76.3%, and 80.7% for the novel classes,

as shown in Table 3. The performance level of the novel classes did not drop significantly compared with the base classes, despite their large difference in the number of training samples, which indicates that the proposed method was effective in transferring the features learned from the base classes to the novel classes. The proposed transformer-based method also achieved an average mAP of 81.7% when using 50-shots for finetuning. Similar patterns were found when using 10-shots and 30-shots for finetuning (see Table 3).

Example results of few-shot detection are shown in Fig. 8. It can be seen from the figure that most target objects are correctly localized with the corresponding object class tags, which indicates the effectiveness of

Table 2
Number of Samples in Testing Dataset.

Object class	Number	Object class	Number	Object class	Number
Aerial_Lift	27	Fence	42	Railing	42
Anchorage	34	Guardrail_System	29	Roof	39
Arm	31	Guardrail	42	Safety_Hook	27
Back	35	Hard_Hat	49	Safety_Net	31
Barricade	42	Harness	34	Scaffold	30
Body_Belt	31	Hole	23	Scaffold_Platform	29
Connector	30	Ladder	30	Scaffold_Pole	44
Coupler	31	Lanyard	30	Step	33
Cover	32	Leading_Edge	34	Toeboard	25
Crane	31	Lifeline	28	Tool	31
Employee	37	Personal_Fall_Arrest_System	26	Trash	34
Excavation	32	Positioning_Device_System	27	Walking_Working_Surface	37
Excavator	41	Post	43	Wall	48

Table 3

Experimental Results of Few-Shot Object Detection.

Experiment	P	mP	R	mR	mF	mAP	average mAP	bP	mbP	bR	mbR	mbF	nP	mnP	nR	mnR	mnF
K = 50-Split1	88.2	88.2	81.0	79.5	83.6	83.9	81.7	88.6	88.6	81.5	80.3	84.2	84.3	85.6	76.8	76.3	80.7
K = 50-Split2	88.1		79.3			80.7		88.2		80.3			87.7		77.2		
K = 50-Split3	88.4		78.2			80.5		88.9		79.0			84.7		75.0		
K = 30-Split1	75.9	75.3	72.2	71.6	73.4	73.1	72.8	76.5	75.9	73.3	72.8	74.4	70.7	69.8	62.6	61.1	65.2
K = 30-Split2	75.2		71.5			72.8		75.9		72.8			69.1		60.1		
K = 30-Split3	74.8		71.2			72.4		75.4		72.4			69.6		60.7		
K = 10-Split1	52.3	51.8	50.3	49.7	50.7	49.2	48.3	53.4	52.9	52.5	51.9	52.4	42.7	42.1	31.1	30.5	35.4
K = 10-Split2	51.6		49.7			48.7		52.7		51.9			42.0		30.5		
K = 10-Split3	51.4		49.2			46.9		52.5		51.4			41.8		30.0		
K = 50-ResNet-Split1	85.1	84.6	72.5	72.1	77.9	75.5	75.1	85.7	85.2	73.2	72.9	78.6	79.8	79.3	66.4	85.1	84.6
K = 50-ResNet-Split2	84.6		72.0			74.9		85.2		72.8			79.3		65.0	84.6	
K = 50-ResNet-Split3	84.1		71.9			74.8		84.7		72.7			78.8		64.9	84.1	
General-all	N/A	85.7	N/A	73.7	79.2	N/A	78.8	N/A	89.5	N/A	78.9	83.9	N/A	33.8	N/A	21.6	26.4
General-balanced set	N/A	60.6	N/A	57.4	59.0	N/A	52.8	N/A	60.9	N/A	59.2	57.6	N/A	60.1	N/A	54.2	57.0

Note: P=Precision (%); mP = mean Precision (%); R = Recall (%); mR = mean Recall (%); mF = mean *F-1* measure (%); mAP = mean average precision; bP=Precision for base classes (%); bR = Recall for base classes (%); nP=Precision for novel classes (%); and nR = Recall for novel classes (%).

**Fig. 8.** Example results of proposed few-shot object detection method.

the proposed method. Fig. 9 shows more such examples and some failure cases for illustrating the error causes. For example, as shown in Fig. 9 (a), the model failed to distinguish regular pedestrians from construction workers on a busy street, because most instances in the dataset are workers, and the model tends to predict a person more often as a construction worker. In addition, since there is no pedestrian object class, no prior knowledge or supervision was provided for the model to make such distinction. As shown in Fig. 9 (b), it is difficult for the model to detect objects that blend into each other (arm, tool, and lanyard in this case), especially when the lightening condition was not satisfactory, because there can be limited visual cues or features (only silhouette in this case) for the model to make an accurate prediction. As shown in Fig. 9 (c), the model also tends to detect mirrored workers or equipment as real object instances, because the mirrored objects can have similar features. In addition, there can exist insufficient number of such samples in the dataset for the model to learn to filter out objects caused by reflection. As shown in Fig. 9 (d), the model can fail to distinguish a baseball cap from a hard hat, because those two objects can have similar colors, shapes, and locations, especially for images taken from a distance or when there are partial occlusions. Most of such failure cases can be

improved if more prior knowledge could be incorporated, thus it might be beneficial to add a few non-fall-related objects such as pedestrians and mirrored tools, or frequent noncompliant objects such as baseball caps, in future work, so that the model can learn to make more accurate predictions.

Ambiguities in object classes is another source of error, despite that the same semantics are maintained between site objects and construction safety regulations. For example, depending on the specific operation or activity a worker is performing, walls or roofs can sometimes be considered as walking/working surfaces as well, which can cause confusion for the model to classify the detected wall or roof objects. Similarly, a dee-ring can be considered as a connector when it is connected with other components of a personal fall arrest system; but, it can be considered as a safety hook when it is used to fasten a safety net. Moreover, one object class can refer to a variety of objects. For example, the tool class can include buckets, screwdrivers, and towels, and the trash class can include debris, wasted materials, plastic bags, and empty food boxes. Considering our dataset size, it may not contain sufficient training samples for each subcategory of tool or trash object (even fewer than 50-shots), thus the performance can be negatively affected. To

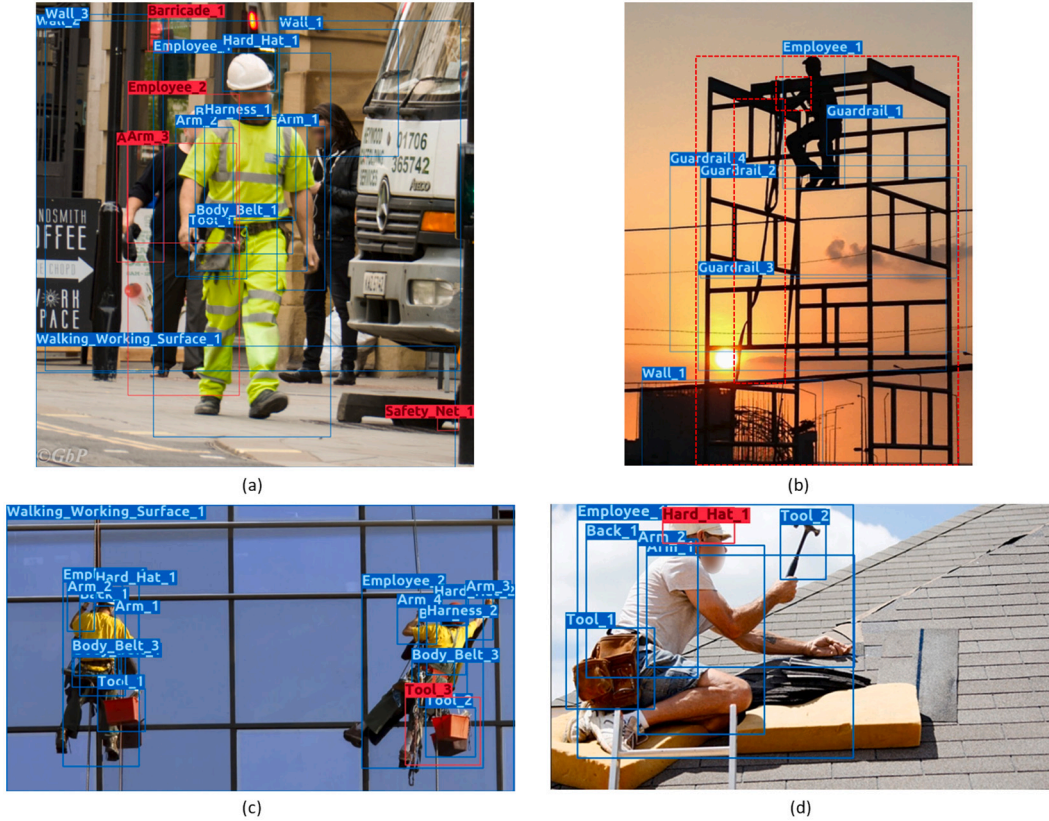


Fig. 9. Examples of failure cases: (a) Errors when there are pedestrians; (b) Errors when objects blend into each other; (c) Error when there is reflection; and (d) Error when there is baseball hat.

further improve the detection performance, in future work, more detailed object subclasses could be considered and more relevant site images could be retrieved.

Errors can also arise when there are no sufficient visual cues. For example, workers can wear harnesses inside their jackets due to cold weather, with only a small part of lanyard outside to connect with a lifeline, in which case it is difficult to determine if they are using a complete set of personal fall arrest system and to further assess its condition. For images taken under foggy weather, site objects can be too blurry to be captured. Those extreme cases are very different from partial occlusions that are frequently seen onsite, which can be effectively solved by taking images from another angle or making inference using other existing features. For those cases, however, measures other than vision-based methods, such as different sensing technologies, need to be incorporated to help obtain useful site information for supporting subsequent compliance comparisons.

5.2.2. Comparative performance evaluation with state-of-the-art methods

The performance of the proposed few-shot detection method was further compared with the performance of a general detection method, the DETR method by Carion et al. [14]. The DETR is a basic encoder-decoder transformer without a set of hand-crafted components (see Section 2.1.1). The comparison of the DETR and the proposed method was conducted under two different settings: (1) using all images, including all base class images and K -shots ($K = 50$) novel class images, which aims to evaluate if the proposed method can help narrow the performance gap between base and novel classes; and (2) using balanced K -shots ($K = 50$) images as described in Section 4.2.2 for all object classes, following the evaluation method in Kim and Chi [38], which aims to evaluate if the proposed method can help improve the performance on novel classes. Thus, the DETR model was first trained using all images. As shown in Table 2, the DETR model achieved an average

precision, recall, and $F-1$ measure of 85.7%, 73.7%, and 79.2%, respectively, for all object classes, which is comparable, or slightly decreased than the proposed model (88.2%, 79.5%, and 83.6%) using 50-shots for finetuning as well. However, such performance was achieved by maximizing the performance of base classes (i.e., 83.9% in $F-1$ measure), while sacrificing the performance of novel classes (i.e., 26.4% in $F-1$ measure). The proposed few-shot detection method, on the other hand, shows balanced performance for all object classes, since the performance difference between the base and novel classes is relatively small (i.e., 84.3% vs. 80.7%).

The DETR model was then trained and tested using the balanced image set. The DETR model achieved an average precision, recall, and $F-1$ measure of 60.6%, 57.4%, and 59.0%, respectively, for all object classes, as per Table 2. Thus, the proposed few-shot detection method shows 27.6% (88.2% vs. 60.6%), 22.1% (79.5% vs. 57.4%), and 24.6% (83.6% vs. 59.0%) improvement in precision, recall, and $F-1$ measure, respectively, when using 50-shots for finetuning, which indicates its effectiveness in learning from only a limited number of site images, and utilizing the knowledge learned from base classes to enhance the overall performance. The proposed few-shot detection method, therefore, shows better potential for generalizability and practicality to apply onsite, given the difficulty in collecting a balanced dataset and the possible severe consequences of failing to detect certain rare site objects.

In addition, this study also compared the performance of the proposed method with a state-of-the-art few shot detection method based on residual network (ResNet) [15], as described in Section 2.1.2. Using 50-shots, the proposed method showed an improvement of 3.6% (88.2% vs. 84.6%) in precision, 7.4% (79.5% vs. 72.1%) in recall, and 5.8% (83.6% vs. 77.9%) in $F-1$ measure for all object classes, which indicates the effectiveness of the proposed method. This may also indicate that, for this domain-specific application/dataset, transformer-based models can achieve better few-shot detection performance compared to ResNet-

based models.

5.3. Attribute recognition

5.3.1. Performance of proposed attribute recognition method

At both, object and attribute levels, the proposed attribute recognition method achieved good performance: an average precision, recall, and *F-1* measure of 94.8%, 95.7%, and 95.3%, respectively, at the object level, and 90.4%, 96.1%, and 92.5% at the attribute level, as shown in Fig. 10. This indicates that the proposed method was effective in localizing and recognizing domain-specific attributes and dealing with imbalanced distributions. The achieved performance is comparable (or even higher) to those reported in other applications outside of the construction domain. For example, an *F-1* measure of 80.2% was achieved on the richly annotated pedestrian (RAP) dataset [19] and 86.85% on the PETA dataset [58].

However, small variations in performance can be observed for different attribute classes, due to four main reasons. First, some attribute classes appear more frequently, which caused the model to better recognize them for achieving the minimum errors. For example, since a large portion of the walking/working surfaces are horizontal, the model performed better in recognizing the attribute of horizontal than vertical. Similarly, a large portion of the equipment are undamaged, therefore the model achieved better performance in recognizing attribute of undamaged than damaged. Second, recognizing some attributes can be easier by nature because it only requires low-level features such as lines, curves, and colors, while recognizing some other attributes can require finer details or implicit information such as the connections between objects. For example, better performance was achieved for recognizing attributes such as damaged, undamaged, level, and not_level, since they require only geometry, shape, or contour information. However, a decrease in the performance can be observed for recognizing attributes such as fixed and not_fixed, because the model needs to zoom into a small area for identifying the connectivity between objects. Similarly, to determine if an edge is protected/unprotected, the model needs to understand the connections between a leading edge and its surroundings, therefore can achieve lower performance in such case. Third, some attributes are difficult to be correctly inferred. For example, for recognizing the attributes steep and low_sloped, the model needs to learn the cut-off ratio of the roofs that were perceived from different viewpoints, thus is difficult to differentiate. Similarly, for the attributes fully_planked and not_fully_planked, it is difficult for the model to understand how large the gap between scaffold boards would be considered neglectable, which can lead to errors in prediction. Fourth, some attributes can require dynamic information to be correctly identified, which is hard to acquire from static site images. For example, it is difficult to distinguish whether the worker is standing or walking due to the lack of motion information, which often occurs through consecutive

images.

Compared to the few-shot object detection results in Section 5.2, the proposed method achieved better performance in recognizing attributes for two main reasons. First, since each object can have multiple attributes as illustrated in Fig. 3, there are significantly more training samples of attributes, especially given the difference in the numbers of object and attribute classes (39 objects classes vs. 21 attribute classes). Second, recognizing attributes can require relatively lower-level features or certain fixed visual cues than detecting objects. For example, the attribute recognition model can use the shape information of a scaffold platform to distinguish if it is damaged or not, as shown in Fig. 3. However, to detect an object as a scaffold board, the object detection model needs to use a set of higher-level features such as its texture, its spatial location and connection with other scaffold components or workers, and its implicit part-whole connections with scaffold, to distinguish it from other related objects. Therefore, better performance was achieved for identifying attributes than objects in this study.

5.3.2. Comparison of proposed method with method without spatial attention localization modules

To evaluate the effectiveness of the proposed attribute recognition method in learning regional features for attributes with different detail levels, the aforementioned results were compared with the performance of the method without the SALMs. Instead, a fully connected layer was used to calculate attribute predictions from the concatenated features. Then the best predictions were selected from predictions at different feature levels using the maximum voting scheme. Such method without SALMs achieved an average precision, recall, and *F-1* measure of 87.7%, 88.9%, and 88.3%, respectively, at the object level, and 80.2%, 74.8%, and 77.5%, respectively, at the attribute level, as shown in Fig. 10. Therefore, the SALMs were able to improve 7.1% (94.8% vs. 87.7%), 6.8% (95.7% vs. 88.9%), and 7.0% (95.3% vs. 88.3%) in precision, recall, and *F-1* measure, respectively, at the object level, as well as 9.9% (90.4% vs. 80.5%), 21.3% (96.1% vs. 74.8%), and 15.0% (92.5% vs. 77.5%) at the attribute level, which indicates its effectiveness in capturing attribute-specific location information and adaptively allocating attention accordingly to attribute detail levels.

6. Limitations

Seven limitations of this study are acknowledged. First, the proposed method focused on detecting fall-related objects and their attributes from site images. Images can be limited in continuously monitoring site conditions compared to videos. In future work, the authors plan to further extend this method to site videos. A similar level of performance is expected since site videos consist of a sequence of related images. Second, the proposed few-shot detection method used SwAV as the backbone model and a basic encoder-decoder transformer for encoding and localizing the objects, without testing on other deep learning model structures. In future work, the authors plan to explore different types of model architectures, such as region similarity representation learning (ReSim) [61] and other transformer-based models, to further improve the detection performance. Third, the proposed few-shot detection method was tested under limited settings: using three different base/novel splits and comparing with only one state-of-the-art general or few-shot detection method. In the future, the authors plan to further test the proposed method by conducting a comprehensive ablation study, in order to fully understand the impact of its components, its robustness under varying conditions, as well as its performance compared to additional state-of-the-art methods. Fourth, the feasibility and performance of the proposed few-shot detection and attribute recognition methods were tested using online site images. In their future work, the authors plan to conduct additional testing on site images from real construction sites (e.g., collected using UAVs), since such images may vary in viewpoints and/or object scales. However, only a small amount of adaptation is expected since the proposed deep learning method

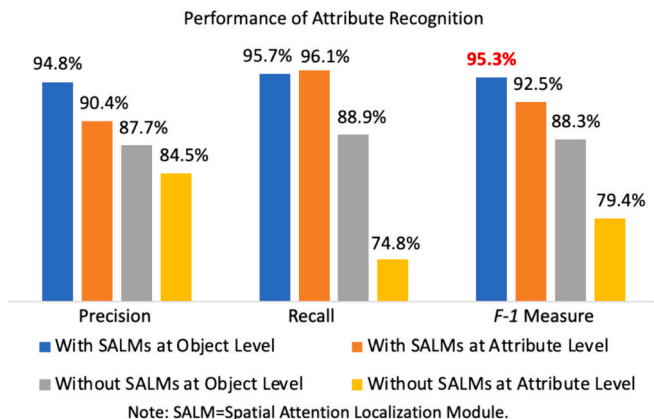


Fig. 10. Performance of attribute recognition model.

already considers generating samples of different viewpoints and scales. Fifth, the proposed method was tested on a fall-related dataset of limited size. In the future, the authors plan to further expand the dataset to include more image examples. As a result, more equipment components, non-fall-related objects, and noncompliant objects need to be considered. Sixth, the proposed attribute recognition method considered a total of 21 frequent attribute classes as described in construction safety regulations, which could be limited as more hazard scenarios are considered. Thus, the authors plan to consider more attribute classes that are less frequent but are also important in determining compliance, such as different types of material that can affect the capacity or strength of an equipment, and eventually the choice of the protection measures. Seventh, this study used two separate deep learning models for object detection and attribute recognition because they were regarded as two different tasks. However, this could slow down the process to collect necessary information for subsequent compliance checking. In future work, the authors plan to conduct joint detection of site objects and their associated attributes to decrease the processing time for identifying such information.

7. Contributions to the body of knowledge

This study offers a deep learning-based method to detect multiple fall-related site objects with imbalanced distribution and recognize their associated attributes for describing site conditions using computer vision techniques. This study improves the existing object detection efforts in the construction safety domain in three primary ways. First, it detects multiple types of fall-related site objects, including granular objects such as equipment components, and recognizes intricate attributes for each detected object to add semantic descriptions about their conditions. Importantly, the detected objects and attributes are following the same semantics as in the construction safety regulations (i. e., the names and meanings of the extracted objects and attributes are matching the entity names and meanings in the regulations). Recognizing different objects and their intricate attributes helps capture important condition information from visual site images to provide essential facts for subsequent compliance checking given various possible hazard scenarios onsite. Following the same semantics as in the regulations means that the extracted objects and attributes are readily mapped to the entities in the regulation. This results in consistent concept representation between the detected site information and the requirements in the regulations, which facilitates subsequent compliance assessment. Second, this research conducts few-shot detection to address the imbalanced distribution of site objects, which shows good generalizability to detect novel classes with a limited number of site images. Such imbalanced distribution reflects the real site conditions and the difficulty in obtaining site images of certain objects. The few-shot learning method proposed in this study can help reduce the number of training samples required for less frequent fall-related objects, by utilizing the rich features captured from more frequent objects. The semi-supervised region proposal generator, especially the unsupervised backbone model, can extract features in a self-supervised manner, thus significantly lessens the labor of acquiring and annotating a large amount of site images. Moreover, if the proposed method is to be adapted to other accident types such as electrocution, it can also reduce the amount of effort in developing a dataset of electrocution-related objects in large quantities. Third, this study utilizes a few features or techniques to enhance few-shot object detection and attribute recognition performance, including relative location information of the objects, the connectivity between them, and the spatial attention localization modules to associate attributes with specific regions and allocate attention according to the detail levels of the attributes. Such features and techniques help efficiently utilize the hidden information from site images, which future research studies could incorporate when exploring other neural network structures.

8. Conclusions and future work

This paper proposed a deep learning-based method to identify fall-related objects, with multiple attributes, to provide intricate site condition information for supporting field compliance checking. The proposed method consists of three submethods: dataset development, few-shot object detection, and attribute recognition. For dataset development, site images covering different fall-hazard scenarios were retrieved from open image resources through an image query process. Labels that are consistent with the entity names from OSHA regulations were added to the collected images. Data augmentation was conducted to add slightly modified copies of the original images for introducing more variations. Then the proposed few-shot object detection method uses a transformer with position encoding and a semi-supervised region proposal generator that extracts composition features to detect novel classes, by leveraging the knowledge learned from detecting base classes. The proposed attribute recognition method conducts multi-label classification for recognizing attributes of the detected objects as semantic descriptions about their conditions (which are consistent with entity names from OSHA regulations), by using spatial attention localization modules to associate attributes with specific regions and allocate attention according to detail levels.

The developed fall-related dataset contains 1,098 images with 39 object classes and 21 attribute classes. The proposed few-shot detection method was tested on the developed dataset with three different splits of base and novel object classes, and achieved an average precision, recall, and *F-1* measure of 88.2%, 79.5%, and 83.6%, respectively, for all object classes, with 88.6%, 80.3%, and 84.2% for base classes and 85.6%, 76.3%, and 80.7% for novel classes, when using 50-shots during fine-tuning, which indicates good few-shot detection performance. It also shows that the features learned from the base classes were effectively transferred to predicting the novel classes. Compared with a state-of-the-art ResNet-based few-shot detection method, the proposed method showed an improvement of 3.6% (88.2% vs. 84.6%) in precision, 7.4% (79.5% vs. 72.1%) in recall, and 5.8% (83.6% vs. 77.9%) in *F-1* measure for all object classes, which indicates the effectiveness of using transformer-based models for this domain-specific application/dataset. Compared with general detection method, the proposed few-shot detection method showed good improvement in increasing the performance of novel classes without sacrificing the performance of base classes. The proposed attribute recognition method achieved an average precision, recall, and *F-1* measure of 94.8%, 95.7%, and 95.3%, respectively, at the object level, as well as 90.4%, 96.1%, and 92.5% at the attribute level. Compared to the method without using spatial attention localization modules, the proposed attribute recognition method showed effectiveness in using attribute-specific location information and varying amounts of attention to enhance performance.

In their future work, the authors plan to extend this research work in three main research directions. First, different deep learning model structures could be selected as the backbone model, encoder block, or decoder block, and tested for evaluating their effectiveness in boosting detection performance. For example, deformable transformers that use multi-scale deformable attention mechanisms and thus show better performance than basic transformers [62] can be adapted, trained, and comparatively evaluated for selecting the most appropriate model architecture. Second, the authors will further evaluate the proposed method using images collected by UAVs from real construction sites, as well as adapt the method to site videos. UAV images can contain slightly different distributions of viewpoints and/or object scales. Using UAV images could, thus, help further evaluate and improve the generalizability of the proposed method, which lays a solid foundation for its real-world application. Utilizing videos can help continuously collect site information for making compliance comparisons, thus can replace the existing expensive labor work of safety managers to monitor site safety. Third, beyond detecting objects and attributes, the authors will develop a visual relation detection method to understand the pairwise

relations between objects with the same relation class names as in [46], as well as to provide a structured representation of all the detected information. Visual relation detection is also an important step for field compliance checking because it enriches the detected object and attribute information by adding more semantic descriptions about their interactions, spatial relations, and restrictions, based on which a site scene can be fully understood. By exploring these three directions, the authors seek to develop better detection methods and underlying models with sufficient accuracy, extract necessary visual information for a variety of scenarios, and use as limited number of site images or videos as possible, for establishing an effective and automated field compliance checking process to enforce compliance consistently with minimum human assistance.

CRedit authorship contribution statement

Xiyu Wang: Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. **Nora El-Gohary:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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